DO NOW — RECALL

Heuristics, modelling and pipelining

Name: _____ Class: _____ Date: _____

1. What does a heuristic provide?

2. What is throughput?

3. What is a pipeline stall?

4. Binary search for 13 in indexes0–14 of [0,1,…,14]. List midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A rule-of-thumb estimate useful for choosing or ordering options.

2. Model answer

Completed work per unit time.

3. Model answer

Delay caused by unavailable stage/result or resource.

4. Model answer

mid7, then mid11, then mid13; found at 13.

By the end, you can…

• Define the heuristic

• Model the trade-off

• Pipeline and performance models

Vocabulary — know these terms

heuristic — Estimate/rule guiding search.

performance model — Simplification estimating resource use.

pipeline — Overlap processing stages.

stall — Delay in progress through stages.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain the method

Task four completes stage S3 in cycle 6, assuming no stalls; it is not completed in cycle 4.

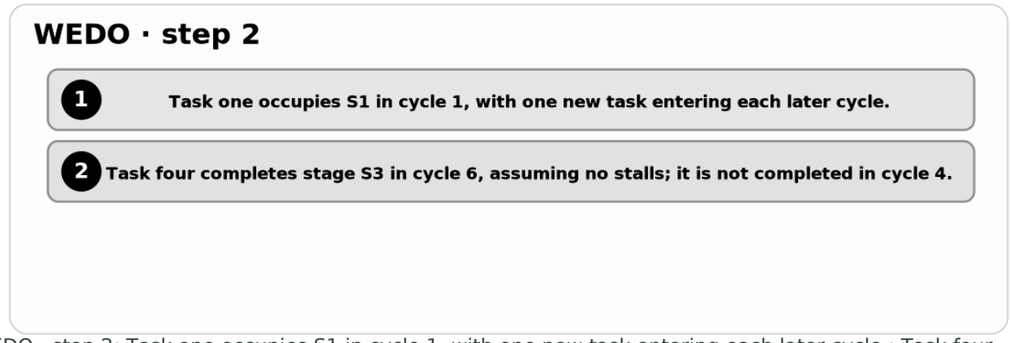
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A three-stage pipeline receives four independent tasks, one task per stage per cycle. When does the fourth fin

Task one occupies S1 in cycle 1, with one new task entering each later cycle.



WEDO · step 2; Task one occupies S1 in cycle 1, with one new task entering each later cycle.; Task four completes stage S3 in cycle 6, assuming no stalls; it is not completed in cycle 4.

NEW LEARNING

Read, annotate and explain

Define the heuristic

A heuristic quickly estimates how promising a choice is; it is not automatically an exact solution.

Model the trade-off

Separate local edge cost, accumulated cost and estimated remaining cost before choosing a path.

Pipeline and performance models

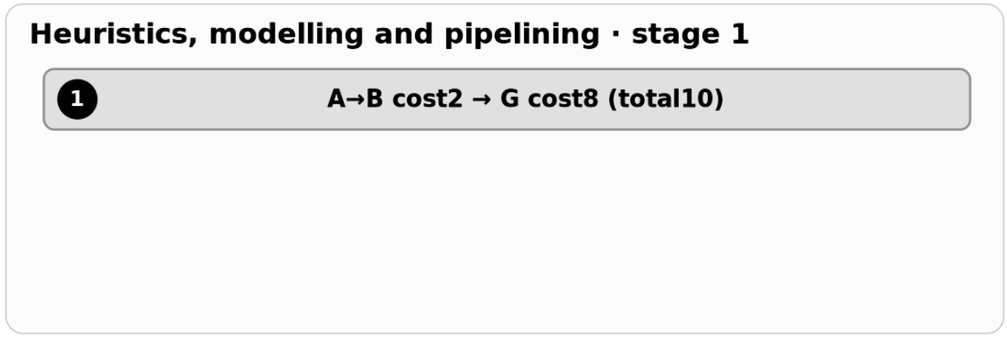
A pipeline can overlap independent stages, increasing throughput, but dependencies and resource constraints may cause stalls.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

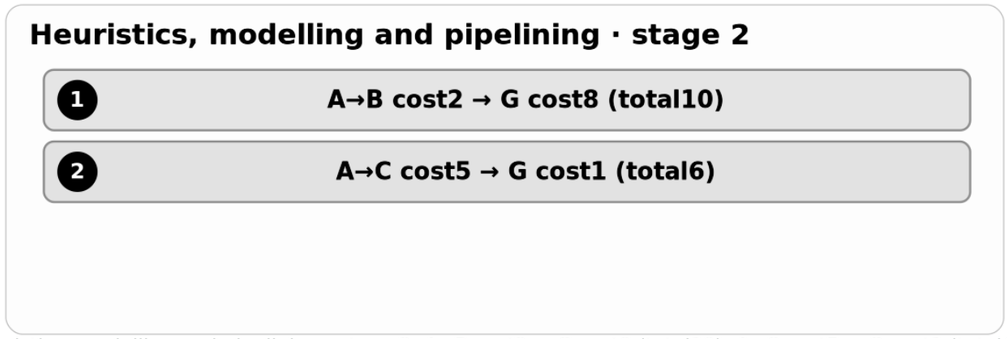
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Heuristics, modelling and pipelining · stage 1; A→B cost2 → G cost8 (total10)

New Learning



Heuristics, modelling and pipelining · stage 2; A→B cost2 → G cost8 (total10); A→C cost5 → G cost1 (total6)

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Choosing the cheapest immediate edge always yields the cheapest complete path.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

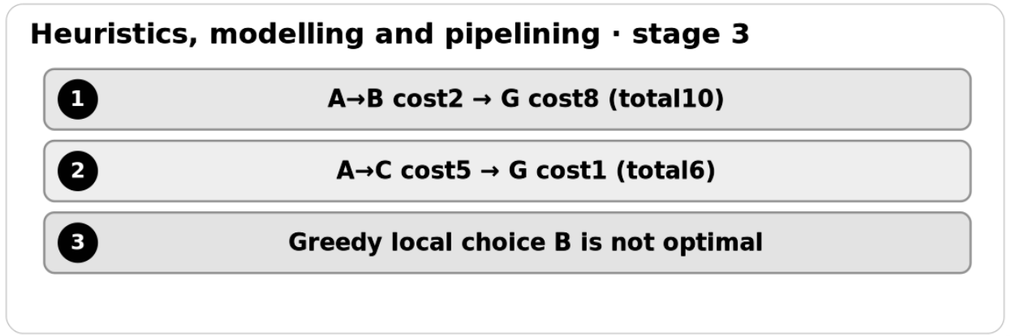
Question 3

Evaluate using an A* heuristic and a computational performance model to plan emergency routes. [6]

New Learning · original lesson diagrams

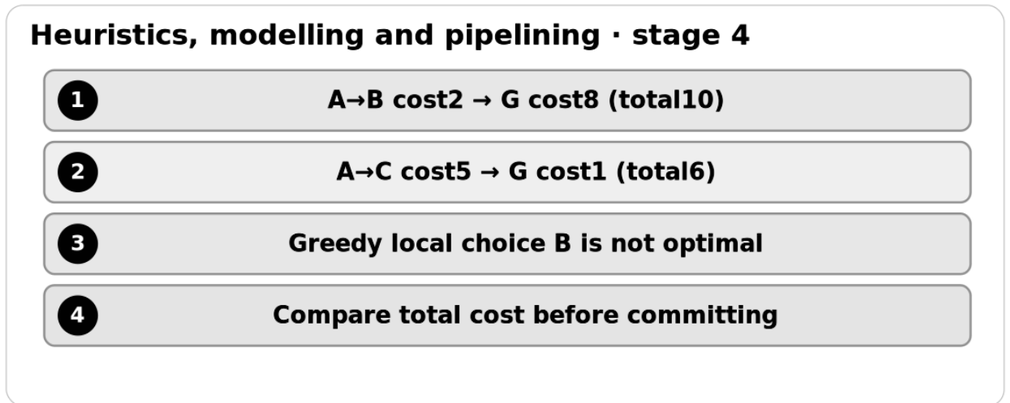
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Heuristics, modelling and pipelining · stage 3; A→B cost2 → G cost8 (total10); A→C cost5 → G cost1 (total6); Greedy local choice B is not optimal

New Learning



Heuristics, modelling and pipelining · stage 4; A→B cost2 → G cost8 (total10); A→C cost5 → G cost1 (total6); Greedy local choice B is not optimal; Compare total cost before committing

I DO — WORKED EXAMPLE

Follow the worked method

Evaluate two routes A–B–G (3+7) and A–C–G (5+1).

Predict first

Calculate complete route costs, rather than deciding only from the first edge.

Explain the method

The via-C path is cheaper at six total units despite a larger first edge.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain why choosing the cheapest immediate edge is not guaranteed to produce a shortest path. [4]

Question 2

Five independent jobs pass through 3 one-cycle stages. Give ideal completion cycle for all five and compare with non-overlapped 15 cycles. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Task one occupies S1 in cycle 1, with one new task entering each later cycle.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

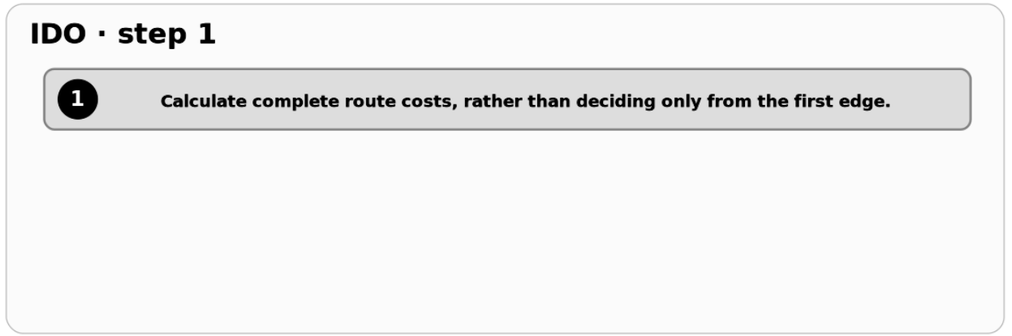
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Evaluate two routes A–B–G (3+7) and A–C–G (5+1).

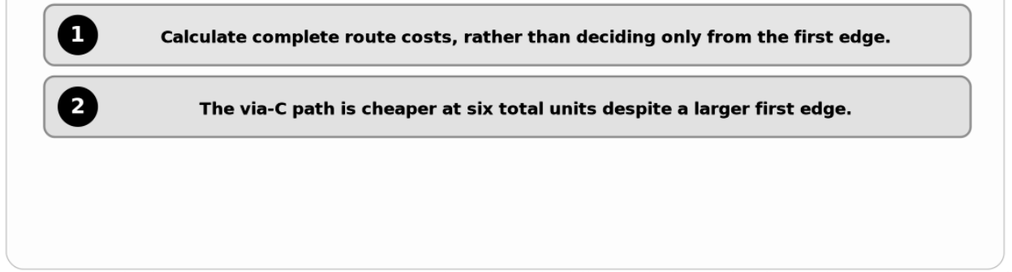
Calculate complete route costs, rather than deciding only from the first edge.



IDO · step 1; Calculate complete route costs, rather than deciding only from the first edge.

Evaluate two routes A–B–G (3+7) and A–C–G (5+1).

Calculate complete route costs, rather than deciding only from the first edge.



IDO · step 2; Calculate complete route costs, rather than deciding only from the first edge.; The via-C path is cheaper at six total units despite a larger first edge.

WE DO — GUIDED PRACTICE

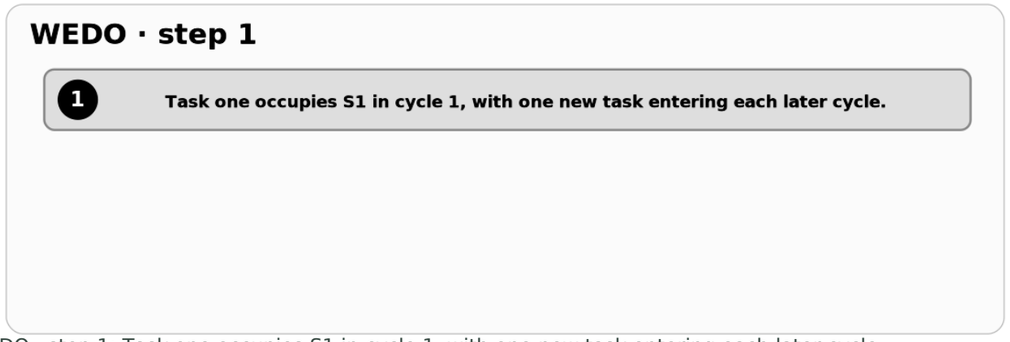
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

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WEDO · step 1; Task one occupies S1 in cycle 1, with one new task entering each later cycle.